WELCOME

TO OUR PRESENTATION

RideGuard SOS

PRESENTED BY

- ➢ ARINDAM NATH (LEADER)
- ➢ ARGHYA DAS
- ➢ NISHANI BHUNIYA
- ➢ RATUL KOLE

Problem Analysis

The Problem

Bike accidents can become more dangerous when the injured rider is unable to call for help.

Key Problems

- ➢ Rider may become unconscious after an accident.
- ➢ Nobody may be nearby to contact family immediately.
- ➢ Emergency contacts may not know the accident location.
- ➢ Manual phone calling may not be possible after a serious crash.
- ➢ Delay in informing family or rescuers can increase risk.

Problem Statement

How can we automatically detect a motorcycle accident and instantly alert emergency contacts with the rider's location when the rider is unable to call for help?

Proposed Solution

Our Solution – RideGuard SOS

RideGuard combines a **smart helmet and mobile application**.

➢ **Working:**

1. Accident occurs
MPU6050 detect sudden acceleration, impact and abnormal movement.

2. Arduino analyzes sensor data
The Arduino checks whether the movement crosses the accident threshold.

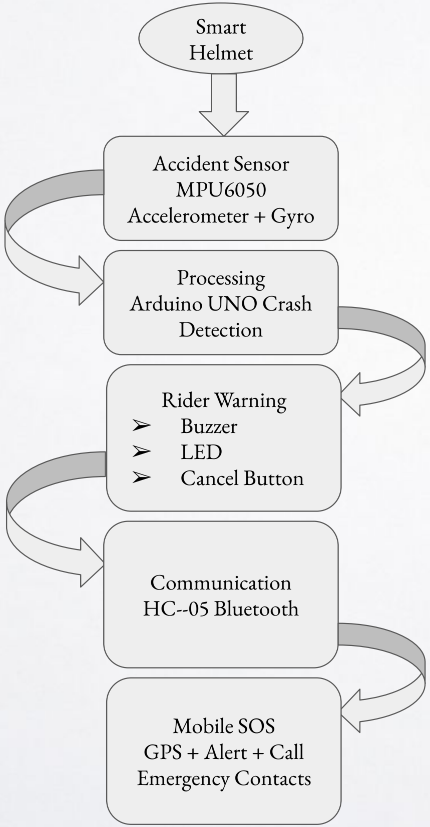
3. Warning starts
Buzzer activates and gives the rider a short time of 7 seconds to cancel the SOS.

4. No response
Arduino sends an accident signal to the phone through Bluetooth.

5. Phone activates SOS
The mobile application gets GPS location and alerts saved emergency contacts.

Important feature: A cancel button prevents false alerts if the helmet is dropped accidentally

Hardware & Software Architecture



Hardware:

- ➢ Arduino Uno
- ➢ MPU6050 accelerometer + gyroscope
- ➢ Buzzer
- ➢ Push Button
- ➢ LEDs
- ➢ Breadboard
- ➢ Jumper Wires
- ➢ Battery for final prototype

Software:

- ➢ Arduino Firmware for sensor monitoring
- ➢ Crash-detection algorithm
- ➢ Bluetooth communication
- ➢ Android application
- ➢ GPS/location service
- ➢ Emergency contact management

Technology Stack

Technologies Used

Embedded Hardware: Arduino Uno, MPU6050, HC-05 Bluetooth

Programming: Arduino IDE, C/C++

Mobile Application: Android Studio, Kotlin or Java

Communication: Bluetooth Serial Communication

Location: Smartphone GPS, Google Maps location link

Future improvements

- ➢ High-G accelerometer for better impact detection
- ➢ Hands-free helmet calling
- ➢ Helmet-wearing detection
- ➢ Cloud-based accident history
- ➢ Nearby hospital/emergency-service integration

➢ Why RideGuard SOS is Needed

❑ INDIA'S ROAD SAFETY CHALLENGE

- **4.8+ lakh** road accidents reported in India in 2023
- **1.72+ lakh** people lost their lives in road accidents
- **Two-wheelers are among the most vulnerable road users**
- **Overspeeding is a major contributor to road accidents**
- After a serious crash, the rider may be **unable to call for help or share their location**

➢ What Happens After an Accident?

Accident Happens → Rider cannot respond → Emergency response is delayed

➢ OUR SOLUTION

Metric	Number
Road Accidents	4,80,583
Fatalities	1,72,890

➢ THE PROBLEM ⚠️

- **After a serious motorcycle accident, the rider may be unable to call for help, and emergency contacts may not know the rider's location**

➢ OUR SOLUTION 🛡️

- **RideGuard SOS** automatically detects a suspected accident, gives the rider a short cancellation window, and sends the rider's GPS location to emergency contacts.

Thank You!